

Pondicherry University 2 marks Question & Answer

1. What does mean by matter waves?(May 2014, May 2017, Nov 2019)

During the movement of particles like electrons, protons, atoms and molecules, should have some type of wave motion associated with them. These waves are called as matter waves.

2. State the de Broglie hypothesis.(Jan 2013, May 2014)

According to de Broglie “the wave-particle dualism should not be restricted to radiation alone, but should also be extended to all the fundamental bodies of physics”

3. What do you mean de Broglie wavelength?(May 2013)

The wavelength of the wave associate with a moving particle is called as de Broglie wavelength. Particle with momentum ‘mv’ will have de Broglie wavelength of $\lambda = h / p$ (or) h / mv

4. What is de Broglie wavelength of an electron of an electron which has been accelerated from rest through a potential difference of 100 V.

(Jan 2011, May 2018, Nov 2019)

We have the relation $\lambda = 12.25 / \sqrt{V} \text{ \AA}$

In this case $V = 100$ volts. Hence $\lambda = 12.25 / \sqrt{100} = 1.225 \text{ \AA}$

5. State Heisenberg’s Uncertainty principle.

(Jan 2011, 2013, 2016, May 2013, 2017, Nov & May 2018, Nov 2019)

The principle states that the exact position and momentum of a particle cannot be determined simultaneously with a desired accuracy.

Taking Δx as the error in determining its position and Δp the error in determining its momentum at the same instant, then $\Delta x \cdot \Delta p = h / 2\pi$

6. Give the physical significance of ” ψ “. (Jan 2015, 2016, May 2018)

Wave function “ ψ ” and its first order partial derivatives with respect to x, y, z must be continuous everywhere. “ ψ ” must be finite for all values of x, y and z . The wave function “ ψ ” beside being normalized must be single valued as $P(t)$ can have only one value at a particular place at a particular time.

7. Give the meaning of “ ψ ” in wave mechanics.(Jan 2014)

The wave function ψ is referred to probability amplitude for the position of the particle. While ψ itself has no physical interpretation the square of its absolute magnitude $|\psi|^2$ evaluate at a particular point at a particular time is proportional to the probability of experimentally finding there at that point.

8. State Tunnel effect.(Jan 2014, 2016, Nov & May 2018, Nov 2019)

If a particle is impinging on a barrier with energy less than the height of the potential barrier; it will not necessarily be totally reflected by the barrier but there is always a probability that it may cross the barrier and continue its forward motion.

9. Write a short note on properties of matter waves. (Jan 2016. May 2017)

- a) Lighter is the particle, greater is the wavelength associated with it.
- b) Smaller is the velocity of the particle, greater is the wavelength associated with it.
- c) If $v = 0$, then $\lambda = \text{infinity}$, i.e., wave become indeterminate and if $v = \text{infinity}$ then $\lambda = 0$. This indicates that the De-Broglie waves are generated by the motion of particle.
- d) These waves are produced in the particles whether the particles are charged particles or uncharged particles (in the equation $\lambda = h/m v$, there is no term representing the charge. It is independent of charge). This shows that these waves are not EM waves but are a new kind of waves.
- e) The velocity of matter wave is not constant since it depends on the velocity of the material particle whereas the velocity of EM wave is constant.
- f) The velocity of these waves is greater than velocity of light. They can travel faster than light.

10. Give the expression for the de-Broglie wavelength of an accelerated electron (May 2019)

If a charged particles, say an electron is accelerated by a potential difference of 'V' volts, then its kinetic energy is given energy by

$$\text{K.E} = e V$$

$$e V = \frac{1}{2} m v^2$$

$$2eV = m v^2$$

Multiply 'm' on both sides

$$2meV = m^2 v^2$$

Take sq. root on both sides

$$mv = \sqrt{2meV} \quad \dots\dots\dots (1)$$

Then, the wavelength of electron is given by

$$\lambda = h/m v \quad \dots\dots\dots (2)$$

Substitute eqn. (1) in (2), $\lambda = \frac{h}{\sqrt{2meV}}$

11. Write down the Schrodinger time independent wave equation (May 2019, Sep 2020)

The Schrodinger time independent wave equation

$$\nabla^2 \Psi + \frac{2m}{\hbar^2} (E - V) \Psi = 0$$

12. Calculate the de-Broglie wavelength associated with a proton moving with a velocity of $1/10^{\text{th}}$ of velocity of light. Given the mass of proton is $1.67 \times 10^{-27} \text{ kg}$.(Jan 2016)

Solution

Mass of the proton $m = 1.67 \times 10^{-27} \text{ kg}$

Velocity of the proton $v = 1/10 \times \text{velocity of light}$

$$= 1/20 \times 3 \times 10^8$$

Velocity of light $v = 3 \times 10^8 \text{ m/s}$

De-Broglie wavelength $\lambda = h/mv$

$$\lambda = 6.626 \times 10^{-34} / 1.675 \times 10^{-27} \times 3 \times 10^7 = 1.3186 \times 10^{-14} \text{ m}$$

De-Broglie wavelength $\lambda = 1.3186 \times 10^{-14} \text{ m}$

13. Explain the term degenerate and Non d degenerate energy level. (Jan 2016, May 2018, Jan 2019)

A set of two or more states having the same energy and different wave function is known as degenerate energy level.

A set of two or more states having the same energy and same wave function is known as non-degenerate energy level

14. Calculate the De-Broglie wavelength of an electron accelerated by a potential difference of 150 volt. (May 2019, Sep 2020)

Given $h = 6.626 \times 10^{-34} \text{ Js}$, $m = 9.1 \times 10^{-31} \text{ kg}$

Charge of electron = $1.6 \times 10^{-19} \text{ C}$

Solution:

$$\begin{aligned} \lambda &= h / \sqrt{2MeV} \\ &= \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 1.6 \times 10^{-19} \times 9.1 \times 10^{-31} \times 150}} \\ \lambda &= 1.227 \times 10^{-10} \text{ m.} \end{aligned}$$

15. Mention the ideas which prompted de Broglie to propose his concept of Matter Waves. (Sep 2020)

i) Nature prefers symmetry. If radiation light can act like wave sometime and like a particle at other time , then material particles (eg: electrons, neutrons etc.) should also act as wave at sometime and particles in some other times.

ii) Our universe is fully composed of light and matter.

iii) Every moving particle is always associated with waves. The particle is controlled by this wave.

16. Calculate the permitted energy levels of an electron in a box of 1 width 1Å (mass of electron m= 9.16x10⁻³¹ kg) (Jan 2016)

Solution

$$\text{permitted electron eneries is given by } = \frac{n^2 h^2}{8ma^2}$$

for the first enery levels , n = 1 and ence wehave ,

$$E_1 = \frac{1^2 (6.6 \times 10^{-34})^2}{8(9.1 \times 10^{-31}) \times (10^{-10})^2}$$

$$E_1 = 38 \text{ eV.}$$

This corresponds to the minimum energy

For n=2,

$$E_2 = \frac{2^2 (6.6 \times 10^{-34})^2}{8(9.1 \times 10^{-31}) \times (10^{-10})^2}$$

$$E_2 = 152 \text{ eV, } E_2 = 4 E_1$$

Similarly,

$$E_3 = 342 \text{ eV} = 9 E_1.$$

PONDICHERRY UNIVERSITY PART B QUESTIONS

1. What are matter waves? Calculate the De- Broglie wavelength of an electron accelerated through a potential difference of 150 volts.

(Jan 2013)

During the movement of particles like electrons, protons, atoms and molecules, should have some type of wave motion associated with them. These waves are called as matter waves.

Given $h=6.626 \times 10^{-34}\text{Js}$, $m= 9.1 \times 10^{-31}\text{kg}$ Charge of electron = $1.6 \times 10^{-19}\text{C}$

Solution:

$$\begin{aligned}\lambda &= h/\sqrt{2MeV} \\ &= \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 1.6 \times 10^{-19} \times 9.1 \times 10^{-31} \times 150}} \\ \lambda &= 1.227 \times 10^{-10}\text{m}\end{aligned}$$

2. Derive the Schrodinger time dependent wave equation and hence obtain the time independent wave equation.

(Jan 2011, Jan 2013, May 2013, May 2012, May 2017, Nov & May 2018, Nov 2019, Sep 2020)

Let us consider a particle of mass m , moving with a velocity ' v '. let ' Ψ ' be the wave function of the particle along x , y and z coordinates axis . The De-Broglie wavelength associated is given by

$$\lambda = \frac{h}{mv}$$

The classical differential equation of progressive wave moving with a wave velocity ' v ' can be written as

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \Psi}{\partial t^2} \dots\dots\dots (1)$$

where

$$\begin{aligned}\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} &= \nabla^2 \text{ (Laplacian operator)} \\ \therefore \nabla^2 \Psi &= \frac{1}{v^2} \frac{\partial^2 \Psi}{\partial t^2} \dots\dots\dots (2)\end{aligned}$$

The solution for equation (2) is assumed to be

$$\Psi = \Psi_0 e^{-i\omega t} \dots\dots\dots (3)$$

Where ' Ψ_0 ' is the amplitude of the wave at the point (x , y , z)

Diff eqn. (3) w.r.t to 't' twice we get

$$\frac{\partial \Psi}{\partial t} = \Psi_0 e^{-i\omega t} \cdot (-i\omega)$$

$$\frac{\partial^2 \Psi}{\partial^2 t} = \Psi_0 e^{-i\omega t} (-i\omega)(-i\omega)$$

$$\frac{\partial^2 \Psi}{\partial^2 t} = i^2 \omega^2 \Psi_0 e^{-i\omega t}$$

Since, $i^2 = -1$ and $\Psi = \Psi e^{-i\omega t}$

$$\frac{\partial^2 \Psi}{\partial^2 t} = -\omega^2 \Psi \quad \dots\dots\dots (4)$$

Substituting (4) in (2) we get

$$\therefore \nabla^2 \Psi = -\frac{\omega^2 \Psi}{v^2}$$

$$\nabla^2 \Psi + \frac{\omega^2}{v^2} \Psi = 0 \quad \dots\dots\dots (5)$$

But, $\omega = 2\pi\nu = 2\pi \frac{v}{\lambda}$ [where $\nu = \frac{v}{\lambda}$]

$$\text{or } \frac{\omega}{v} = \frac{2\pi}{\lambda} \quad \dots\dots\dots (6)$$

Squaring eqn. (6) on both sides, we get

$$\text{or } \frac{\omega^2}{v^2} = \frac{4\pi^2}{\lambda^2} \quad \dots\dots\dots (7)$$

Substituting (7) in (5), we get

$$\nabla^2 \Psi + \frac{4\pi^2}{\lambda^2} \Psi = 0 \quad \dots\dots\dots (8)$$

By De-Broglie's equation we know $\lambda = \frac{h}{mv}$

$$\therefore \nabla^2 \Psi + \frac{4\pi^2 m^2 v^2}{h^2} \Psi = 0 \quad \dots\dots\dots (9)$$

If 'E' is the total energy of the particle, 'V' is the potential energy and $\frac{1}{2}mv^2$ the kinetic energy, then

$$\text{Total energy } E = \text{P.E} + \text{K.E}$$

$$E = V + \frac{1}{2}mv^2$$

$$E - V = \frac{1}{2}mv^2$$

$$2(E - V) = mv^2$$

Multiplying by 'm' on both sides in above equation we get,

$$2m(E - V) = m^2 v^2 \quad \dots\dots\dots (10)$$

Substituting (10) in (9) we get

$$\nabla^2 \Psi + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0 \quad \dots\dots\dots (11)$$

The above equation is known as **Schrödinger Time Independent wave equation**

For a one-dimensional motion, the Schrödinger equation takes the form

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0$$

Introducing $\hbar = \frac{h}{2\pi}$ we get

$$\nabla^2 \Psi + \frac{2m}{\hbar^2} (E - V) \Psi = 0 \quad \dots\dots\dots (12)$$

For free particle $V=0$, therefore Schrödinger wave equation becomes.

$$\nabla^2 \Psi + \frac{2mE}{\hbar^2} \Psi = 0 \quad \dots\dots\dots (13)$$

4.7.2. Time Dependent wave Equation

We can easily derive Schrodinger time dependent wave equation by eliminating E from Schrodinger time independent wave equation.

Differentiating the Eq. (14) with respect to time t, we get.

$$\Psi(x,y,z) = \Psi_o(x,y,z) e^{-i\omega t} \quad \dots\dots\dots (14)$$

$$\frac{\partial \Psi}{\partial t} = -i \omega \Psi_o e^{-i\omega t} \quad \dots\dots\dots (15)$$

$$= -i (2\pi\nu) \Psi_o e^{-i\omega t} \quad (\because \omega = 2\pi\nu)$$

$$\frac{\partial \Psi}{\partial t} = -i (2\pi\nu) \Psi_o e^{-i\omega t} \quad \dots\dots\dots (16)$$

$$= -i (2\pi\nu) \Psi$$

$$= -2\pi i \frac{E \Psi}{h}$$

$$(\because E = h \nu \text{ then } \frac{E}{h} = \nu) ; (\frac{h}{2\pi} = \hbar)$$

$$\therefore \frac{\partial \Psi}{\partial t} = -i \frac{E \Psi}{\hbar} \quad \dots\dots\dots (17)$$

Multiplying i on the both sides,

$$i \frac{\partial \Psi}{\partial t} = -i \times i \frac{E \Psi}{\hbar}$$

$$(\because i \times i = -1)$$

$$i \frac{\partial \Psi}{\partial t} = \frac{E \Psi}{\hbar}$$

$$\text{or} \quad E \Psi = i\hbar \frac{\partial \Psi}{\partial t} \quad \dots\dots\dots (18)$$

Substitute, The value of $E\Psi$ from Eq. (18) in Schrodinger time independent wave Eq. (14) we have

$$\begin{aligned}\nabla^2\Psi + \frac{2m(E-V)\Psi}{\hbar^2} &= 0 \\ \nabla^2\Psi + \frac{2m(E\Psi-V\Psi)}{\hbar^2} &= 0 \\ \nabla^2\Psi + \frac{2m(\hbar\frac{\partial\Psi}{\partial t} - V\Psi)}{\hbar^2} &= 0 \\ \nabla^2\Psi &= -\frac{2m(\hbar\frac{\partial\Psi}{\partial t} - V\Psi)}{\hbar^2} \\ \frac{-\hbar^2}{2m}\nabla^2\Psi &= \hbar\frac{\partial\Psi}{\partial t} - V\Psi \\ \frac{-\hbar^2}{2m}\nabla^2\Psi + V\Psi &= \hbar\frac{\partial\Psi}{\partial t} \quad \dots\dots\dots (19)\end{aligned}$$

The equation is known as **Schrodinger time dependent** wave equation. Eq. (19) may be written as

$$\begin{aligned}\left(\frac{-\hbar^2}{2m}\nabla^2 + V\right)\Psi &= \hbar\frac{\partial\Psi}{\partial t} \\ H\Psi &= E\Psi \quad \dots\dots\dots (20)\end{aligned}$$

Where, $H = \left(\frac{-\hbar^2}{2m}\nabla^2 + V\right)$ is Hamiltonian operator and $E = \hbar\frac{\partial}{\partial t}$ is the energy operator

3. **Explain the application of Schrodinger equation using one dimensional potential problem. Solve it to obtain eigen functions and show that eigen values are discrete. (May 2012, Jan 2014, 2015, 2016, May 2017, 2018)**

Application of Schrödinger's equation to one dimensional box : (or) particle in a one-dimensional box.

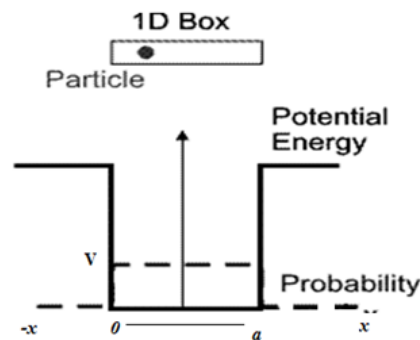


Fig:4.2. Particle in One Dimensional box

Let us consider a particle (e^-) of mass ' m ' moving along ' x -axis' enclosed in '1-D' potential box, since the walls are of potential the particle does not penetrate out the box. The particle is confined between the length ' a ' of the box and has elastic collisions with the walls. Therefore the potential energy of the electron inside the box is constant and can be taken as zero. We can say that outside the box and on the wall of the box, the potential energy ' V ' of the electron is infinity. Inside the box the potential energy ' (V) ' of the electron is zero.

To find the wave functions of the particle within the box of length 'l' we will take the 1-D'

$$\text{Schrodinger time independent wave equation (i.e) } \frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2} (E-V) \Psi = 0 \quad \dots\dots\dots (1)$$

Since the potential inside the box is zero [(i.e)V=0) the particle has kinetic energy alone and thus it is named as a free particle (or) free electron.

∴ For a free particle, the Schrodinger wave Eq. is given by

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2} E \Psi = 0 \quad \dots\dots\dots (2)$$

$$\frac{d^2\Psi}{dx^2} + k^2\Psi = 0 \quad \dots\dots\dots (3)$$

$$k^2 = \frac{2m}{\hbar^2} E \quad \dots\dots\dots (4)$$

Eqn. (3) is a second order differential equation. Therefore it should have solution with two arbitrary constants. The solution of Eqn. (3) is given by

$$\Psi(x) = A \sin kx + B \cos kx \quad \dots\dots\dots (5)$$

A, B are arbitrary constants, these constants can be found out by applying boundary conditions.

(i.e) $V(x) = \infty$. When 'x = 0' and 'x = 'a'

Applying Boundary conditions:

(i) At $x = 0$; Potential Energy $V = \infty$; there is no chance for finding the particle at the walls of the box.

$$\Psi(x) = 0$$

Applying these conditions in equation (5) we get

$$0 = A \sin 0 + B \cos 0$$

$$0 = 0 + B \quad (1)$$

$$B = 0 \quad \dots\dots\dots (6)$$

(ii) At $x = a$, P.E $V = \infty$;

∴ There is no chance for finding the particle at the walls of the box.

$$\Psi(x) = 0$$

Eq.(5) becomes

$$0 = A \sin ka + B \cos ka$$

From (6); $B = 0$

$$0 = A \sin ka$$

$$\sin ka = 0 ; \text{ since } A \neq 0$$

$$ka = n\pi$$

$$k = \frac{n\pi}{a}, \text{ where, } n=0,1,2,3 \quad \dots\dots\dots (7)$$

Substituting Eqn. (6) and Eqn. (7) in Eqn. (5) we get

$$\Psi(x) = A \sin \frac{n \pi x}{a} \dots\dots\dots (8)$$

Energy of the Particle

From Eqn. (4)

$$k^2 = \frac{2mE}{\hbar^2} = \frac{2mE}{\frac{\hbar^2}{4\pi^2}}$$

$$k^2 = \frac{8\pi^2 E m}{h^2} \dots\dots\dots (9)$$

Squaring Eqn. (7) we get

$$k^2 = \frac{n^2 \pi^2}{a^2} \dots\dots\dots (10)$$

Equating the Eqn. (9) and (10) we get

$$\frac{8\pi^2 E m}{h^2} = \frac{n^2 \pi^2}{a^2}$$

Energy of the particle $E_n = \frac{n^2 h^2}{8ma^2} \dots\dots\dots (11)$

From Eqn.(11) we can say that there are ‘n’ number of energy levels and corresponding wave function. Thus we can say that each value of ‘E_n’ is known as “Eigen value” and the corresponding value of ‘Ψ_n’ is called Eigen function.

Energy levels of an Electron

For various values of ‘n’ we get various energy values of the electron. The lowest energy value (or) ground state energy value can be got by substituting n = 1 in Eqn.(11)

When n=1 ; we get; $E_1 = \frac{h^2}{8ma^2}$

$$n=2 ; \text{ we get; } E_2 = \frac{4h^2}{8ma^2} = 4E_1$$

$$n=3 ; \text{ we get; } E_3 = \frac{9h^2}{8ma^2} = 9E_1$$

$$n=4 ; \text{ we get; } E_3 = \frac{16h^2}{8ma^2} = 16E_1$$

In general, $E_n = n^2 E_1 \dots\dots\dots (12)$

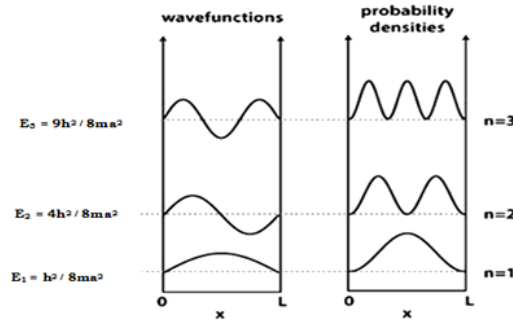


Fig.4.3. Energy levels of an electron

It is found from the energy levels $E_1, E_2, E_3, \dots$ the energy levels of an ' Ψ_n ' are discrete from the various energy levels mentioned above and it can be clearly said that quantum mechanics deals with discrete energy levels.

Normalization of the wave function:

It is the process by which the probability (P) of finding the particle (ie) inside the box can be done.

$$P = \int_0^a |\Psi|^2 dx = 1 \quad \dots\dots\dots (13)$$

For, a '1 - D' Box traveling in the 'x-direction' with the length 'a'.

Substitute Eqn. (8) in Eqn.(13) we get,

$$P = \int_0^a A^2 \sin^2 \frac{n \pi x}{a} dx = 1$$

$$A^2 \int_0^a \left| \frac{1 - \cos \frac{2n \pi x}{a}}{2} \right| dx = 1$$

$$A^2 \left[\frac{x}{2} - \frac{1}{2} \left[\frac{\sin \frac{2n \pi x}{a}}{\frac{2n \pi}{a}} \right] \right]_0^a = 1$$

[Since the second term of the integral becomes zero at both limits]

$$A^2 = \left[\frac{x}{2} \right]_0^a = 1, \text{ Thus, } \frac{A^2 a}{2} = 1$$

$$A^2 = \frac{2}{a} \quad (\text{or}) \quad A = \sqrt{\frac{2}{a}} \quad \dots\dots\dots (14)$$

Substituting the value of 'A' in Eqn. (8) we get the normalized wave function as

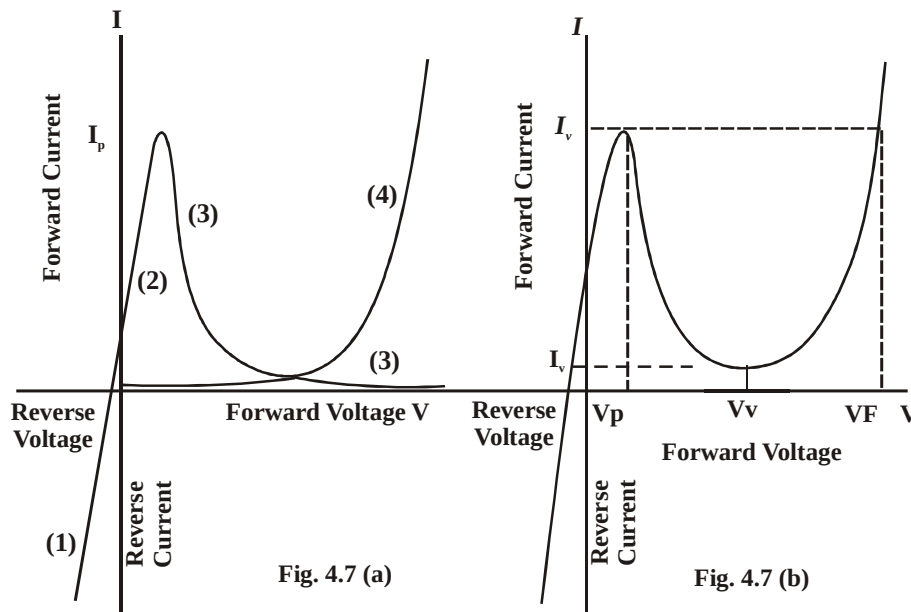
$$\Psi(n) = \sqrt{\frac{2}{a}} \sin \frac{n \pi x}{a} \quad \dots\dots\dots (15)$$

4. Explain in detail the basic principle, construction and operation of a Tunnel Diode.(May 2014, Jan 2015, May 2017, Nov 2018, Nov 2019, Sep 2020)

Draw I-V characteristics of a tunnel diode under forward bias and reverse bias and explain it with the help of energy band diagram.(Jan 2016).

The volt –ampere characteristic of Tunnel Diode:

The following figure shows the volt –ampere characteristic of a tunnel diode under Reverse Bias and Forward Bias condition.



Reverse Bias Characteristic:

Let us consider that the p material is grounded and that a voltage applied across the diode, shifts the n-side with respect to the p-side. If a reverse-bias voltage is applied the n-side level must shift downward with respect to the p-side levels as indicated in the below figure 4.8.

We now observe that there are some occupied energy states [the dark shaded region] in figure 4.8 in the valence band of the p-side which lie at the same level as allowed empty states in the conduction band on the n-side. Hence these electrons will tunnel from the p to the n-side, giving rise to a reverse diode current. As the magnitude of the reverse bias increases, the heavily shaded dark area grows in size, causing the reverse current to increase as shown by number (2) of figure 4.7 (a).

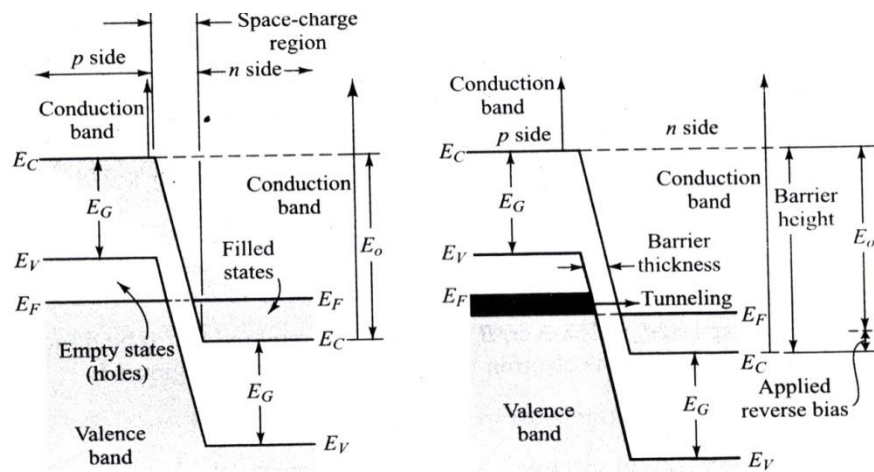
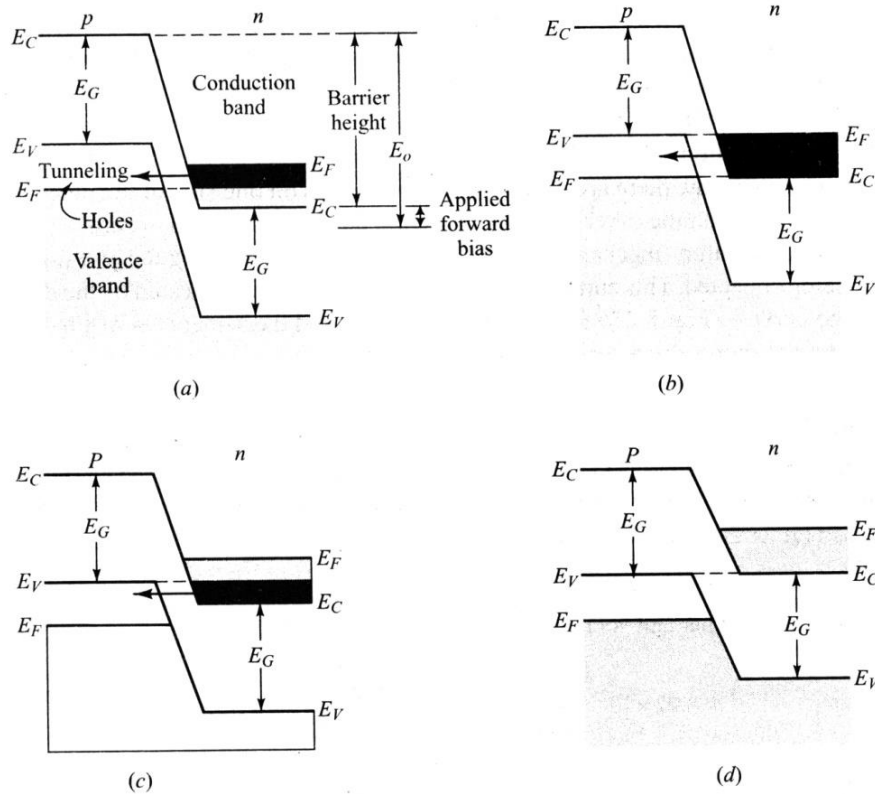


Fig. 4.8

Forward bias characteristics of tunnel diode

The following figures show the energy band structures of tunnel diode under forward bias nature. When forward bias voltage is applied to the diode the n-side levels must shift upward with respect to those on the p-side, and the energy-band picture for this situation is indicated in figure 4.9 (a). It is now evident that there are occupied states the conduction band of the n material (the heavily shaded dark levels) which are at the same energy allowed empty state (holes) in the valence band of a p-side. Hence electrons will tunnel from the n-side to the p material, giving rise to the forward current as indicated by the number 2 in figure 4.7 (a).

As the forward bias is increased further the condition shown in figure 4.9 (b) is reached. Now the maximum number of electrons can leave occupies states on the right side of the junction and turn through the barrier to empty states on the left side, giving rise to the peak current I_p and the corresponding voltage is called as peak voltage V_p as show in figure 4.7 (a). If still more forward bias is applied the situation in figure 4.9 (c) is obtained, and the tunnels current decreases, to a minimum called as valley current (I_v) and the corresponding voltage is called as valley voltage (V_v) which is shown as number 3 in figure 4.7 (a).



Finally, at an even larger forward bias, the band structure of figure 4.9 (d) is valid. Since now there are no empty allowed states on one side of the junction at the same energy as occupied states on the other side. The tunneling current must drop to zero.

In addition to the quantum-mechanical current described above, the regular p-n junction injection current is also being collected. This current is indicated by the dashed line number as 4 in figure 4.7(a). The figure 4.7 (b) is the sum of the solid and dashed curves of tunnel diode characters.

5. Obtain an expression for the reflection and transmission coefficient for tunneling through a barrier of definite potential (May 2019)

Consider a potential barrier as shown in figure 4.4. In the region $x = 0$ to $x = a$ ($V > E$). According to classical mechanics, a steam of particles coming from the left would therefore be reflected. However, consider a potential barrier as shown in above figure 4.4.

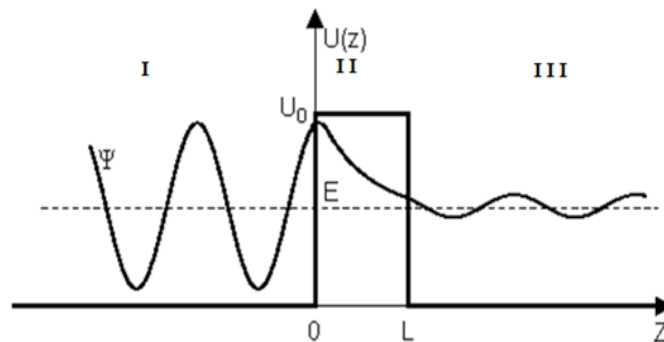


Fig.4.4. Penetration of a Potential barrier

In the region $x = 0$ to $x = a$ ($V > E$). According to classical mechanics, a stream of particles coming from the left would therefore be reflected. However, because of the wave nature of matter, there is some probability that the particle penetrates out of the other side of the barrier as we shall show, it can actually escape into a region $x > a$, where $E > V$. If the particle is impinging on the barrier with energy certainly less than the height of the potential barrier, it will not necessarily be reflected but there is always a probability that it may cross the barrier and continue its forward motion. This probability of crossing the barrier is called the **Tunnel effect**.

The Schrodinger's wave equation is

$$\frac{\partial^2 \Psi_1(x)}{\partial x^2} + \frac{2m(E-Vx)\Psi_1}{\hbar^2} = 0 \quad \dots\dots\dots (1)$$

The Schrodinger's wave equations for different region are: For the region I substitute $V(x) = 0$ in Eqn. (1) we get,

$$\frac{\partial^2 \Psi_1(x)}{\partial x^2} + \frac{2mE\Psi_1}{\hbar^2} = 0 \quad \dots\dots\dots (2)$$

$$\frac{\partial^2 \Psi_1(x)}{\partial x^2} + \Psi_1 k^2 = 0 \quad \dots\dots\dots (3)$$

$$k^2 = \frac{2mE}{\hbar^2} \quad \dots\dots\dots (4)$$

The solution for Eqn. (3) is given by

$$\Psi_1 = Ae^{ikx} + Be^{-ikx} \quad \dots\dots\dots (5)$$

For the region II substitute $V(x) = V_0$ in Eqn. (1) we get,

$$\frac{\partial^2 \Psi_2(x)}{\partial x^2} + \frac{2m(E-V_0)\Psi_2}{\hbar^2} = 0$$

$$\frac{\partial^2 \Psi_2(x)}{\partial x^2} - \frac{2m(V_0-E)\Psi_2}{\hbar^2} = 0 \quad \dots\dots\dots (6)$$

$$\frac{\partial^2 \Psi_2(x)}{\partial x^2} - \beta^2 \Psi_2 = 0 \quad \dots\dots\dots (7)$$

$$\beta^2 = \frac{2m(V_0-E)\Psi_2}{\hbar^2} \quad \dots\dots\dots (8)$$

The solution for Eqn. (7) is given by

$$\Psi_2 = Ce^{-\beta x} + De^{\beta x} \quad \dots\dots\dots (9)$$

For region III substitute $V(x) = 0$ in Eqn. (1) we get,

$$\frac{\partial^2 \Psi_3(x)}{\partial x^2} + \frac{2mE\Psi_3}{\hbar^2} = 0 \quad \dots\dots\dots (10)$$

$$\frac{\partial^2 \Psi_3(x)}{\partial x^2} + k^2 \Psi_3 = 0 \quad \dots\dots\dots (11)$$

$$k^2 = \frac{2mE}{\hbar^2} \quad \dots\dots\dots (12)$$

The solution for Eqn.(12) is given below,

$$\Psi_3 = Ge^{ikx} + He^{-ikx} \dots\dots\dots (13)$$

As there is no reflection in region III, the term He^{-ikx} is discarded.

$$\Psi_3 = Ge^{ikx} \dots\dots\dots (14)$$

Here A, B, C, D and G are the constants whose values can be obtained by applying the following boundary conditions:

In order that Ψ and $\frac{\partial \Psi}{\partial x}$ be continuous at $x=0$, we have

$$(\Psi_1)_{x=0} = (\Psi_2)_{x=0} \dots\dots\dots (15)$$

Applying the condition of Eqn.(15) and Eqn.(5) and Eqn.(9) and substituting $x=0$ we get.

$$A+B = C+D \dots\dots\dots (16)$$

$$\left(\frac{\partial \Psi_1}{\partial x}\right)_{x=0} = \left(\frac{\partial \Psi_2}{\partial x}\right)_{x=\infty} \dots\dots\dots (17)$$

Applying the condition of Eqn.(17) and equating Eqn.(5) and Eqn.(9) and substituting $x=0$ we get.

$$Ae^{ikx} + Be^{ikx} = Ce^{-\beta x} + De^{-\beta x}$$

$$\text{ie., } ikA - ikB = \beta C - \beta D \dots\dots\dots (18)$$

$$ik(A-B) = \beta(C-D) \dots\dots\dots (19)$$

$$(A-B) = \frac{\beta(C+D)}{ik} \dots\dots\dots (20)$$

Adding Eqn. (16) from Eqn. (19) we get

$$2A = \left(1 + \frac{\beta}{ik}\right) C + \left(1 - \frac{\beta}{ik}\right) D \dots\dots\dots (21)$$

Subtracting Eqn. (16) from (19) we get

$$2B = \left(1 - \frac{\beta}{ik}\right) C + \left(1 + \frac{\beta}{ik}\right) D \dots\dots\dots (22)$$

Again in order that Ψ and $\partial \Psi / \partial x$ be continuous at $x=a$, we have,

$$(\Psi_2)_{x=a} = (\Psi_3)_{x=a} \dots\dots\dots (23)$$

Applying the condition of Eqn. (22) and equating Eqn. (19) and Eqn.(14) and substituting $x = a$ we get

$$Ce^{-\beta x} + De^{-\beta x} = Ge^{ikx} \dots\dots\dots (24)$$

$$\left(\frac{\partial \Psi_2}{\partial x}\right)_{x=a} = \left(\frac{\partial \Psi_3}{\partial x}\right)_{x=a} \dots\dots\dots (25)$$

Applying the condition of Eq. (24) and equating Eq.(19) and Eq.(14) and substituting $x=a$ we get

$$\beta C e^{\beta a} - D \beta e^{-\beta a} = ik G e^{ika} \dots\dots\dots (26)$$

$$\beta(C e^{\beta a} - D e^{-\beta a}) = ik G e^{ika}$$

$$(C e^{\beta a} - D e^{-\beta a}) = \frac{ik G e^{ika}}{\beta} \dots\dots\dots (27)$$

Adding Eq.(24) and Eq.(25) we get,

$$C = \frac{1}{2} \left(1 + \frac{ik}{\beta} \right) e^{-\beta a} \cdot Ge^{ika} \quad \dots\dots\dots (28)$$

$$D = \frac{1}{2} \left(1 - \frac{ik}{\beta} \right) e^{\beta a} \cdot Ge^{ika} \quad \dots\dots\dots (29)$$

To find the transmission Coefficient

$$2A = \frac{1}{2} \left(1 + \frac{ik}{\beta} \right) e^{-\beta a} \cdot Ge^{ika} \left(1 + \frac{ik}{\beta} \right) + \frac{1}{2} \left(1 - \frac{ik}{\beta} \right) e^{\beta a} \cdot Ge^{ika} \left(1 - \frac{ik}{\beta} \right)$$

$$2A = \frac{Ge^{ika}}{2} \left[e^{-\beta a} \left[2 + \frac{\beta}{ik} + \frac{ik}{\beta} \right] + e^{\beta a} \left[2 - \frac{\beta}{ik} - \frac{ik}{\beta} \right] \right]$$

$$A = \frac{Ge^{ika}}{4} \left[2e^{-\beta a} + \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] e^{-\beta a} + 2e^{\beta a} - \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] e^{\beta a} \right]$$

$$A = \frac{Ge^{ika}}{4} \left[2(e^{\beta a} + e^{-\beta a}) + \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] (e^{-\beta a} - e^{\beta a}) \right]$$

$$A = \frac{Ge^{ika}}{4} \left[2(e^{\beta a} + e^{-\beta a}) - \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] (e^{\beta a} - e^{-\beta a}) \right]$$

$$A = Ge^{ika} \left[\frac{2(e^{\beta a} + e^{-\beta a})}{4} - \frac{1}{2} \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] \frac{(e^{\beta a} - e^{-\beta a})}{2} \right]$$

The formula for finding transmission coefficient $\frac{1}{T} = \left| \frac{A}{G} \right| \left| \frac{A}{G} \right|^*$

After simplification, the value of T is as follows

$$\frac{1}{T} = \left[1 + \left(\frac{V_0^2}{4E(V_0 - E)} \right) \sinh^2 \beta a \right] \text{ and } T = \left[1 + \left(\frac{V_0^2}{4E(V_0 - E)} \right) \sinh^2 \beta a \right]^{-1}$$

To find reflection coefficient

The formula for finding reflection coefficient $R = \left| \frac{B}{G} \right| \left| \frac{B}{G} \right|^*$

Where,

$$B = \frac{1}{2} \left(1 - \frac{ik}{\beta} \right) \left(\frac{1}{2} \left(1 + \frac{ik}{\beta} \right) e^{-\beta a} \cdot Ge^{ika} \right) + \left(1 + \frac{\beta}{ik} \right) \left(\frac{1}{2} \left(1 - \frac{\beta}{ik} \right) e^{\beta a} \cdot Ge^{ika} \right)$$

$$2B = \frac{Ge^{ika}}{2} e^{-\beta a} \left[\left[\frac{ik}{\beta} - \frac{\beta}{ik} \right] + \frac{1}{2} e^{\beta a} \cdot Ge^{ika} \left[\frac{\beta}{ik} - \frac{ik}{\beta} \right] \right]$$

$$2B = \frac{i}{2} Ge^{ika} e^{-\beta a} \left[\left[\frac{ik}{\beta} + \frac{\beta}{ik} \right] + \frac{i}{2} e^{\beta a} \cdot Ge^{ika} \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] \right]$$

$$B = \frac{i}{4} Ge^{ika} e^{-\beta a} \left[\left[\frac{ik}{\beta} + \frac{\beta}{ik} \right] + \frac{i}{4} e^{\beta a} \cdot Ge^{ika} \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] \right]$$

$$B = \frac{i}{4} Ge^{ika} (e^{\beta a} \cdot e^{-\beta a}) \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right]$$

$$B = -\frac{i}{2} Ge^{ika} \frac{(e^{\beta a} - e^{-\beta a})}{2} \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right]$$

Reflection coefficient, $R = \left| \frac{B}{G} \right| \left| \frac{B}{G} \right|^*$

After simplification, $R = \left[\frac{1}{4} \left(\frac{V_0^2}{E(V_0 - E)} \right) \sinh^2 \beta a \right]^{-1}$